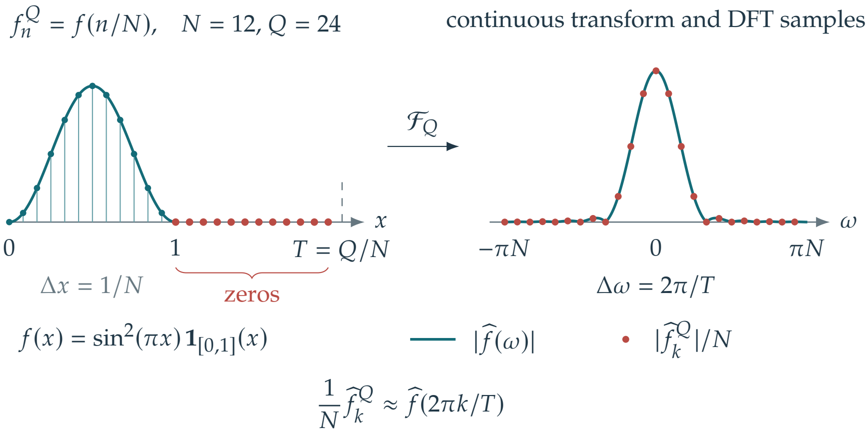

$f_n^Q = f(n/N), \quad N = 12, Q = 24$
continuous transform and DFT samples
$\mathcal{F}_Q$
x
0
1
$T = Q/N$
$\Delta x = 1/N$
zeros
ω
$-\pi N$
0
πN
$\Delta\omega = 2\pi/T$
$f(x) = \sin^2(\pi x)\,\mathbf{1}_{[0,1]}(x)$
$|\widehat{f}(\omega)|$
$|\widehat{f}_k^Q|/N$
$\frac{1}{N}\widehat{f}_k^Q \approx \widehat{f}(2\pi k/T)$